Full Stack Assignment – 13

Assignment ID: FSD-13
Title: Internal Polling Management System
Duration: 3 Hours

1️⃣ Objective

Design and develop an **Internal Polling Management System** that allows users to participate in polls while administrators manage poll creation and results.
The system must enforce voting constraints and ensure **data persistence using a database**.

2️⃣ User Roles

USER

- Register and log in to the system
- View active polls
- Cast a vote in a poll
- View own voting status

ADMIN

- Log in to the system
- Create and manage polls
- View poll results

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Poll Management

- ADMIN can create polls with multiple options
- Poll data must be stored in the database
- Only active polls should be visible to users

Voting

- USER can cast a vote in a poll
- Vote data must be stored in the database

- Poll results must be calculated dynamically

4️⃣ Business Rule (Critical)

⚠️ **A user is allowed to vote only once per poll.**

Any additional voting attempts by the same user must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All poll and vote data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for USER and ADMIN
- Poll listing and voting interface for USER
- Poll creation and result view for ADMIN

UI styling is not evaluated; logic and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions

- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **constraint enforcement, role control, and end-to-end integration**.